



RB001 Week 4 — English Worksheet

Topic: Conditionals — the Magic Door · **Course:** Obby Architect · **Time:** about 45 minutes

This week you learned **if / else**. The code asks a question — if the answer is true, one branch runs; if not, the **else** branch runs. You used this to build a magic door that opens only for players.

1 · Predict 🧠

Read each script. Before you run anything, write what you think will happen.

```
local door = script.Parent

local function onTouch(otherPart)
    local humanoid = otherPart.Parent:FindFirstChildWhichIsA("Humanoid")
    if humanoid then
        door.Transparency = 0.8
        door.CanCollide = false
    end
end

door.Touched:Connect(onTouch)
```

A player touches the door — what happens? A rolling ball touches the door — what happens?


```

local door = script.Parent

local function onTouch(otherPart)
    local humanoid = otherPart.Parent:FindFirstChildWhichIsA("Humanoid")
    if humanoid then
        print("Welcome, player!")
    else
        print("You are not a player.")
    end
end

door.Touched:Connect(onTouch)

```

A loose ball rolls into the door. Which message prints? Why?


```

local password = "magic"

if password == "magic" then
    print("The door opens")
else
    print("The door stays shut")
end

```

Which line prints? What prints if you change the first line to `local password = "abc" ?`

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The script should print **The door opens** when the password is **"magic"**. Instead there is a red error. Count the equals signs.

```
local password = "magic"

if password = "magic" then
    print("The door opens")
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The door should open when a player touches it. There is a red error as soon as the game starts. One word is missing.

```
local door = script.Parent

local function onTouch(otherPart)
    local humanoid = otherPart.Parent:FindFirstChildWhichIsA("Humanoid")
    if humanoid then
        door.Transparency = 0.8
        door.CanCollide = false
    end
end

door.Touched:Connect(onTouch)
```

Hint: every `if ... then` and every `function` needs its own `end` .

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The door should **open** for players (see-through, walk through) and stay solid for everything else. Right now it does the opposite. The two branches are swapped.

```
local door = script.Parent

local function onTouch(otherPart)
    local humanoid = otherPart.Parent:FindFirstChildWhichIsA("Humanoid")
    if humanoid then
        door.Transparency = 0
        door.CanCollide = true
    else
        door.Transparency = 0.8
        door.CanCollide = false
    end
end

door.Touched:Connect(onTouch)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your own obby in Roblox Studio. Build a magic door — when a player touches it, it turns see-through (`Transparency = 0.8`) and lets them walk through (`CanCollide = false`). When you finish, come back here.

Send a photo or video of your magic door, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My **if** checks whether ...

When the condition is true, the door ...

The `else` branch runs when ...

One tricky moment was when ...

To fix it, I ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while a player walks through your magic door. Talk through it like you are teaching someone who has never seen it. Try to use these words: **if**, **else**, **condition**, **true**, **Humanoid**.

1. Show your magic door in Roblox Studio, then run the game.
2. Walk into the door and show it opening for you.
3. Open your script and read the `if humanoid then` line out loud.
4. Say in your own words what **if / else** means.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.